AMIRTESH RAGHURAM

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Vellore Institute of Technology, Vellore

Education

B.Tech Biotechnology

Vellore Institute of Technology, Vellore

2023 - 2027 (Expected) | CGPA: 9.01/10

Professional Experience

Summer Research Intern

Strand Life Sciences, Bangalore | May 2025 - July 2025

- Worked on "Identification of Homologous Regions for the Purpose of Variant Verification in NGS Tests"
- Developed automated Bash scripts for NGS data processing pipelines and homology detection workflows
- Utilized BLAST command-line suite for sequence alignment, variant verification, and homologous region identification
- Implemented bioinformatics solutions for high-throughput genomic data analysis and quality control

First-Author Publications

- Raghuram A. "Computational characterization of Paraphamide B as a potential EGFR-preferential dual EGFR/ALK-binding candidate for non-small cell lung cancer." *In Silico Pharmacology*, Springer Nature, 14, 271. **Published**, 2026.
- Raghuram A. et al. "Computational Design and Validation of a Multi-Epitope Subunit Vaccine Against the Neglected Parasite *Blastocystis* sp." *In Silico Pharmacology*, Springer Nature. **Published**, 2026.

Ongoing Research Projects

AI-AMP Suite: Antimicrobial Peptide Classification and De Novo Peptide Generation (Manuscript in Submission)

- Developed a multi-label XGBoost spectrum classifier predicting Gram-positive, Gram-negative, and antifungal activity from peptide sequence, using ESM2 embeddings and biophysical features selected per task on 14,817 curated sequences from DBAASP, dbAMP, and DRAMP
- Built a binary AMP-versus-non-AMP gatekeeper classifier achieving 0.947 internal AUROC and 0.912 external AUROC on the LMPred dataset (1:14.2 class imbalance), matching or exceeding published classifiers ampir and amPEPpy

- Introduced an adversarial hard-negative stress test using 585 experimentally confirmed-inactive peptides from DBAASP; pre-trained a custom autoregressive transformer (SmallPeptideGPT, 4.77M parameters) on 3.37M PeptideAtlas sequences with 93% generation novelty
- Released all models, training code, and benchmark data via GitHub (github.com/amirtesh/AI-AMP) and deployed a hosted web interface at ppytorch-ai-amp-webapp.ms.fun/app

AD-MTDL: Multi-Task D-MPNN Classifier and ChemProp–REINVENT4 Integration Bridge for Multi-Target Alzheimer's Disease Drug Design (Manuscript in Submission)

- Trained a multi-task directed message-passing neural network (D-MPNN) classifier in ChemProp v2.2.3 on four Alzheimer's disease targets (AChE, BuChE, BACE1, MAO-B) using 11,297 curated ChEMBL 36 compounds with scaffold-balanced splitting and class-weighted BCE loss
- Achieved scaffold-split internal test performance of AUROC 0.853–0.986 and MCC 0.535–0.884 across targets; validated on an independent, near-duplicate-filtered PubChem BioAssay external set of 2,753 compound-target pairs (AUROC 0.728–0.981)
- Developed a lightweight HTTP/Flask scoring bridge decoupling ChemProp v2.x and REINVENT4 into version-isolated environments communicating over localhost, resolving the dependency incompatibility without modifying either codebase
- Conducted five 10,000-step reinforcement learning generation campaigns including a dual-target AChE+BACE1 run (215,065 unique valid SMILES; 74.3% dual-activity concordance among top compounds); performed UMAP chemical space analysis characterizing active-adjacent generation and applicability domain boundaries (github.com/Amirtesh/AD-MTDL)

KinaseForge: Generative AI for Kinase Inhibitor Discovery (Manuscript in Submission)

- Fine-tuned NovoMolGen-157M on kinase-like inhibitor datasets to generate 2.8 million novel kinase-like molecules
- Built XGBoost prediction models for 25 kinases to score and rank the generated compounds
- Created a searchable database enabling multi-target queries — filtering compounds active against specified targets and inactive against others
- Platform accepts custom SMILES strings to predict activity profiles across all 25 kinases; deployed at kinaseforge.vit.ac.in

Multi-Target Inhibitors for Triple Negative Breast Cancer Treatment (Manuscript in Submission)

- Independent research screening natural drug-like compounds as multi-target inhibitors for TNBC therapeutic targets
- Performed triplicate molecular dynamics simulations with MMGBSA binding free energy calculations for robust validation
- Evaluated lead compounds for multi-target efficacy using computational screening and pharmacokinetic profiling
- Assessment of compound stability, binding interactions, and drug-likeness properties

Ferroptosis Predictor: Interpretable QSAR Framework (Manuscript in Submission)

- Developed an interpretable QSAR classification framework distinguishing ferroptosis inducers from inhibitors using a curated, balanced dataset of 1,624 compounds
- Engineered 2,092 molecular descriptors and Morgan fingerprints with scaffold-based cross-validation, achieving AUROC 0.9175 and MCC 0.6617
- Applied SHAP analysis to identify 75 key predictive features; translated discriminative fingerprint signals into chemically meaningful functional group enrichment motifs
- Achieved 86.6% alignment when operationalizing discovered patterns as heuristic screening rules applied to drug-like ChEMBL compounds

Natural Plant-Based Inhibitors for Diabetes Treatment (Manuscript in Submission)

- Collaborative research with PhD scholar identifying natural plant compounds as alpha-amylase inhibitors
- Implementation of computational screening methods, molecular docking, and binding affinity calculations
- Molecular dynamics simulations to evaluate stability of protein-ligand complexes
- Identification of lead compounds with potential therapeutic applications through ADMET profiling

Natural Inhibitors for Glaucoma Treatment (Manuscript in Submission)

- Independent research identifying plant-based inhibitors of human carbonic anhydrase II
- Development of computational pipeline for high-throughput virtual screening
- In-depth molecular dynamics simulations to evaluate stability and effectiveness
- Characterization of binding mechanisms and inhibitory potential of lead compounds

Software Contributions

- **DynaMune:** Ensemble-based protein dynamics analysis platform implementing ENM/NMA, PRS, domain–hinge detection and interface stability analysis using ProDy (github.com/Amirtesh/DynaMune)
- **Torchify:** PyTorch utility library; 4,100+ downloads on PyPI with steady active monthly user adoption. (github.com/Amirtesh/Pytorch-Torchify)
- **Automated-Virtual-Screening:** Binary executable automating ligand screening using Vina, Smina, and QVina with features for file conversion, parallel docking, and results management (github.com/Amirtesh/Automated-Virtual-Screening)
- **Pose-Rescorer:** Physics-based post-docking ligand rescoring workflow using deterministic MM/GBSA with Rapid Perturbation Sampling (RPS) for robustness analysis; validated across EGFR kinase, HIV-1 protease, BRD4 bromodomain, and HSP90 systems (amirtesh.github.io/Pose-Rescorer)
- **QUEST:** CLI toolkit for evaluating ligand conformational strain energies using GFN2-xTB semiempirical QM; integrates tautomer enumeration (RDKit), pocket detection (P2Rank), and molecular docking (Vina) into an automated end-to-end pipeline GitHub
- **Blood-Brain Barrier Permeability Predictor:** ChemProp v2 MPNN-based BBB permeability classifier trained on the B3DB dataset (7,807 compounds); augmented with 7 RDKit physicochemical descriptors (z-scored); MCC-optimised classification threshold (0.58); achieves >0.9 across key validation and test performance metrics — App

Technical Skills

Quantum Chemistry & Biophysics

- Density Functional Theory (ORCA)
- Semi-Empirical QM (xTB)
- Ligand Optimization & Strain Energy
- Conformer & Tautomer Ensembles (CREST)

Programming & Data Science

- Python (NumPy, Pandas, Scikit-learn, BioPython)
- R (dplyr, ggplot2, Bio3D)
- PyTorch & TensorFlow
- Matplotlib/Seaborn
- Statistical Analysis & Machine Learning
- Bash Scripting & Workflow Automation

Structural Bioinformatics

- Protein Structure Analysis (ProDy, Bio3D)
- Protein-Ligand Docking (AutoDock Vina, Smina, QVina, PyRx, AMDock)
- Molecular Dynamics Simulations (GROMACS)
- Binding Free Energy (MM/GBSA, MM/PBSA, gmx_MMPBSA)
- Protein Structure Prediction (AlphaFold2, AlphaFold3, SWISS-MODEL, I-TASSER, Phyre2)
- Structure-Based Drug Design (SBDD)
- Structural Visualization (PyMOL, UCSF Chimera, Discovery Studio)
- Cheminformatics (RDKit, SMILES, QSAR Modeling, Morgan Fingerprints)
- Virtual Screening & Pharmacophore Modeling

Genomics & Next-Generation Sequencing

- RNA-Seq & Differential Expression Analysis
- Single-Cell RNA-Seq (scRNA-Seq, Seurat)
- ChIP-Seq Analysis
- Variant Calling (GATK) & Genome Assembly
- DNA Methylation Analysis (Bisulfite Seq)
- GWAS (PLINK)
- Sequence Alignment (BLAST, BWA, STAR)
- Galaxy, HOMER, BioConductor

Cancer Genomics

- Somatic Mutation Analysis (maftools, oncoplot, lollipopPlot)
- Tumor Heterogeneity and Clonality Assessment
- Mutation Signature Analysis (COSMIC signatures)
- Co-occurrence and Mutual Exclusivity Analysis
- VAF-based Clonal Population Identification
- Oncogenic Driver Identification (Oncodrive)
- rainfallPlot for Mutation Visualization

Computational Drug Discovery & ADMET

- Structure-Based Drug Design (SBDD)
- Virtual Screening & Pharmacophore Modeling
- In Silico Vaccine Design (Multi-Epitope)
- QSAR Modeling (Morgan Fingerprints, Descriptors)
- ADMET Profiling (SwissADME, pkCSM, ProTox 3.0)
- SMILES-based Library Curation & Filtering
- Swiss Target Prediction, MolSoft, KEGG Pathway Analysis

Core Competencies & Achievements

- Quantum-mechanical ligand preparation, including conformer/tautomer ensemble generation and strain energy validation using ORCA and xTB-CREST
- Complete computational drug discovery pipelines from virtual screening to molecular dynamics simulations and binding free energy calculations
- Next-generation sequencing data analysis and interpretation including variant calling and clinical genomics
- Cancer genomics data analysis including somatic mutation profiling, tumor evolution assessment, and clonality studies
- Protein structural analysis, comparative modeling, and structure-function relationship investigations
- Development of custom computational workflows for high-throughput data processing and automation
- Implementation of machine learning approaches for biological data analysis and predictive modeling

Bioinformatics Tools & Resources

ORCA, xTB, CREST, AutoDock Vina, Smina, QVina, GROMACS, gmx_MMPBSA, RDKit, SMILES, AlphaFold2, AlphaFold3, SWISS-MODEL, I-TASSER, Phyre2, PyMOL, UCSF Chimera, Discovery Studio, ProDy, Bio3D, SwissADME, pkCSM, ProTox 3.0, Swiss Target Prediction, SwissDock, MolSoft, KEGG, BLAST, GATK, Galaxy, HOMER, BioConductor, maftools, oncoplot, Oncodrive, rainfallPlot